

Lab Assignment 9

CS 331: Machine Learning

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To be done individually.

The purpose of this lab is to visualize the probabilities that show up in uniform convergence and weak law of large numbers. Let $nSet = \{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$.

1. Write a function to generate a binary classification dataset S_n with n instances. You may take equal class probabilities and $\mathcal{X} \subset \mathbb{R}^2$. Several calls will need to be made to this function.
2. Fix a set $\mathcal{H}$ of 5 hyperplanes of the form $h = (\mathbf{w}, b)$ (the values of co-efficient vector $\mathbf{w}$ and constant b can be generated randomly or can be hard-coded).
3. Write a function that takes a dataset S_n and hypothesis h and computes $\hat{R}_n(h, S_n)$. [This value was referred to as $\hat{R}_n(h)$ in class.]
4. For each h in $\mathcal{H}$, simulate and obtain $R(h)$. To do this you need to generate several datasets $S_n^{(d)}$ of large size n , and obtain the average empirical 0-1 loss obtained by h . For the rest of the assignment treat this value as $R(h)$, $R(h) \approx \sum_{d=1}^D \hat{R}_n(h, S_n^{(d)})/D$.
5. For each h in $\mathcal{H}$ and for each $n \in nSet$ simulate the probability $\mathbb{P}(|\hat{R}_n(h) - R(h)| \geq \epsilon)$.
6. For each $n \in nSet$ simulate the probability $\mathbb{P}(\sup_{h \in \mathcal{H}} |\hat{R}_n(h) - R(h)| \geq \epsilon)$. To do this, for each n you need to draw several datasets S_n . If $|\hat{R}_n(h, S_n) - R(h)| \geq \epsilon$ for some $h \in \mathcal{H}$, such an S_n should count.
So the probability is to be approximated by $\sum_{d=1}^D \frac{\mathbb{1}_{\{\sup_{h \in \mathcal{H}} |\hat{R}_n(h, S_n^{(d)}) - R(h)| \geq \epsilon\}}}{D}$
7. For a fixed value of $\epsilon = 0.1$, generate a single plot of the probabilities in (5) and (6) as a function of n . You should have a curve for each $h \in \mathcal{H}$, and one curve for uniform convergence in (6), so total of 6 curves in a single plot.
8. For a generated training set of n points S_n , obtain $\hat{h}_n^*$ and h^* .

Things to look out for:

1. In question 7, does any curve dominate?
2. Use the plot in (7) to reason if uniform convergence implies weak law of large numbers: Check the N beyond which the probability for UC is less than a fixed value δ . Does this same N end up being a threshold on n for other h for the same δ ? What about the other direction - does the N that work for each h for a δ work for UC probability?
3. Explore (7) with different values of ϵ .
4. Does $\hat{R}_n(\hat{h}_n^*)$ approach $R(h^*)$ with large n ? Try with a finite but larger set $\mathcal{H}$.